PSOC™ Edge E84 training manual for security introduction

About this document

Scope and purpose

This document is a training manual for getting started with PSOC™ Edge E84 MCU security using ModusToolbox™.

Intended audience

The intended audiences for this document are design engineers, technicians, and developers of electronic systems.

Table of contents

About this document **1**
Table of contents **2**
1 Introduction **3**
2 Required development tools **4**
2.1 KIT_PSE84_EVAL details 5
3 Prerequisites **6**
4 Code example: Basic secure application **7**
4.1 Objective 7
4.2 Description 7
4.2.1 Chain of trust with Extended Boot 7
4.2.2 Chain of trust with Extended Boot and edge protect bootloader 8
4.3 Hardware diagram 9
4.4 Project creation 10
4.5 Output 13
4.6 Ownership transfer 14
4.6.1 Create Certificate Signing Request (CSR) 15
4.6.2 Get the CSR signed from Infineon 15
4.6.3 Provision device with your certificate and keys 17
4.7 Enable Secure Boot in Extended Boot 18
4.8 Enable signing of the first user application (cm33 s project) 18
4.9 Conclusion 19
4.10 Challenge 20
4.11 Solution 20
4.12 Revert to default configuration. 22
5 Prerequisites ..... Error! Bookmark not defined.
6 Code example: Basic secure application ..... Error! Bookmark not defined.
6.1 Objective ..... **Error! Bookmark not defined.**
6.2 Description ..... **Error! Bookmark not defined.**
6.2.1 Chain of trust with Extended Boot ..... **Error! Bookmark not defined.**
Revision history **23**
Disclaimer **24**

1 Introduction

This manual provides specific instructions to create, configure, build, and run the application code examples on the PSOC™ Edge E84.

2 Required development tools

- ModusToolbox™ software v3.6 or later
  - Recommended installation via ModusToolbox™ Setup tool
- Eclipse IDE for ModusToolbox™ v2025.4.0 or later
  - Recommended installation via ModusToolbox™ setup tool
- Edge Protect Security Suite v1.6 or later
  - Installed by ModusToolbox™ setup tool as a dependency to ModusToolbox™ v3.6
- ModusToolbox™ Programming Tools v1.5.0 or later
  - Installed by ModusToolbox™ setup tool as a dependency to ModusToolbox™ v3.6
- Board support package (BSP):
  - KIT_PSE84_EVAL_EPC2: v1.0.0 or later (available with ModusToolbox™ v3.6)
- Terminal emulator
  - Instructions in this document use Tera Term
- PSOC™ Edge E84 Evaluation Kit (KIT_PSE84_EVAL)
  - Ensure device boots from external memory:
    - BOOT SW/SW6 in 'ON/HIGH' position (#33 in)
  - Disable alternate serial interface configuration:
    - J20/J21 in Tristate/Not connected (NC) position (#8 in)

Required development tools

2.1 KIT_PSE84_EVAL details

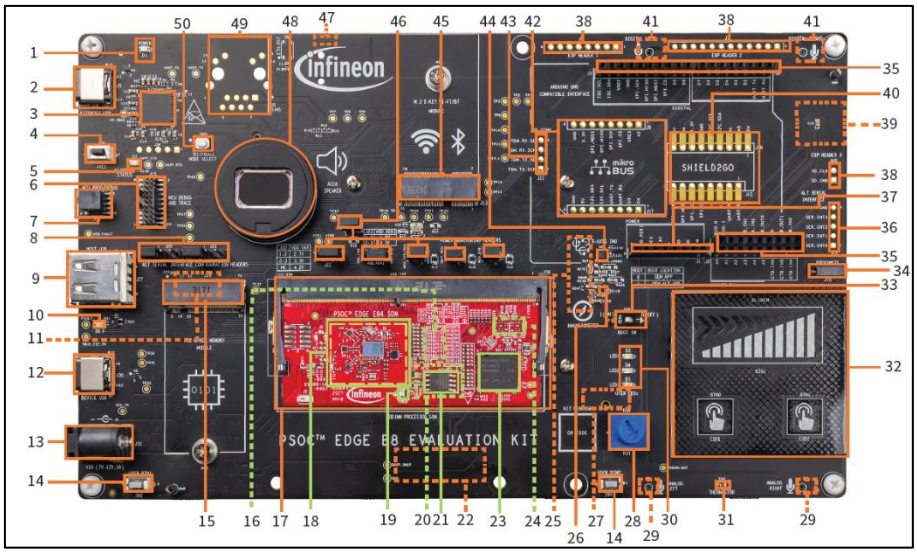

Figure 1 PSOC™ Edge EVK details

25. Baseboard power LED (D1)
26. KitProg3 program/debug USB-C connector (J8)
27. PSOC™ 5LP-based KitProg3 programmer and debugger (CY8C5868LTI-LP039, U2)
28. Reset button (SW1)
29. KitProg3 status LED (D2)
30. PSOC™ Edge E84 MCU ETM/JTAG debug and trace header (J15)
31. PSOC™ Edge E84 MCU 10-pin SWD/JTAG program and debug header (J16)
32. Alternative serial interface configuration headers (J20, J21)
33. PSOC™ Edge E84 MCU USB host Type-A connector (J27)
34. USB-C power delivery (PD) fault LED (D6)
35. Custom display capacitive touch panel connector (J37)**
36. PSOC™ Edge E84 MCU USB device Type-C connector (J30)
37. External power supply VIN connector (J31)
38. PSOC™ Edge E84 MCU user buttons (SW2, SW4)
39. M.2 (B-key) memory interface connector (J29)
40. 128 Mbit Octal-SPI HYPERRAM™ (S70KS1283GABHI020, U12)***
41. Processor system-on-module (SoM) 260-pin SODIMM connector (J28)
42. CYW55513 tri-band (Wi-Fi & Bluetooth®) combo radio (U3)
43. Processor System on Module (SoM) power LED (D3)
44. 1 Gb Octal-SPI NOR flash (S28HS01GTGZBH1030, U10)***
45. 128 Mbit Quad-SPI NOR flash (S25FS128SAGMFB100, U11)
46. MIPI-DSI custom display connector (J38)**
47. PSOC™ Edge E84 MCU (PSE846GPS2DBZC4A, U1)
48. PSOC™ 4000T CAPSENSE™ Co-processor (U9)***
49. Raspberry Pi compatible MIPI-DSI display connector (J39)**
50. 3-axis magnetometers (U4)

1. Raspberry Pi compatible display capacitive touch connector (J41)**
2. Linear potentiometer (R34)
3. Analog microphones (IM73A135V01XTSA1, U36 and U37)**
4. User LEDs (D3, D4, D5)
5. Thermistor (TH1)
6. CAPSENSE™ buttons and slider (CSB1, CSB2, CSS1)
7. BOOT configuration switch (SW6)
8. Proximity sense connector (J19)
9. I/O headers compatible with Arduino UNO R3 (J2, J3, J4)
10. Alternative serial interface I/O header (J14)*
11. Power header compatible with Arduino UNO R3 (J1)
12. PSOC™ Edge E84 MCU expansion I/O headers (J6, J7, J40)*
13. MicroSD card holder (J35)**
14. Infineon's Shield2Go interface headers (J10, J12)*
15. PDM microphones (IM72D128V01XTMA1, U7 and U8)**
16. mikroBUS compatible headers by Mikroelektronika (J9, J17)*
17. Extended I2S header (J11)*
18. 6-axis accelerometer and gyroscope IMU (U5)
19. M.2 (E-key) radio interface connector (J13)
20. PSOC™ Edge E84 MCU power selection/ monitoring headers (J18, J22, J23, J24, J25, J26)
21. Headphone connector (J34)*
22. Speaker (ACC6)
23. RJ45 ethernet magjack connector (J5)*
24. KitProg3 programming mode selection button (SW3)

*Footprint only, not populated on the board
**Component at the bottom side of the Baseboard
***Component at the bottom side of the SoM

3 Prerequisites

Do the following:

1. Install the software and obtain hardware listed in the Required Development Tools section.
2. This document will not cover basic concepts of ModusToolbox™ and PSOC™ Edge. For an introduction to PSOC™, including a getting started guide to ModusToolbox™, go to https://www.infineon.com/product-information/psocdeveloper
3. For PSOC™ Edge trainings, from beginner tutorials to advanced trainings, go to: PSOC™ Edge E84 Training Collection

4 Code example: Basic secure application

4.1 Objective

This code example guides the user to sign a basic application (proj_cm33_s) and activate secure boot in the Extended Boot. With secure boot enabled, the Extended Boot ensures that the application is validated before being loaded.

This example can also be utilised with the Edge Protect Bootloader code example. Once the secured boot is enabled in the Extended Boot, the Edge Protect Bootloader is validated by the Extended Boot, and subsequently, the Edge Protect Bootloader verifies this application and launches it. All projects (proj_cm33_s, proj_cm33_ns, and proj_cm55) need to be signed for the Edge Protect Bootloader to validate the image successfully.

4.2 Description

The code example demonstrates the chain of trust and secured boot flow with “secured boot” enabled or disabled in the Extended boot

4.2.1 Chain of trust with Extended Boot

The chain of trust (CoT) with the Extended boot is as described in Figure 2.

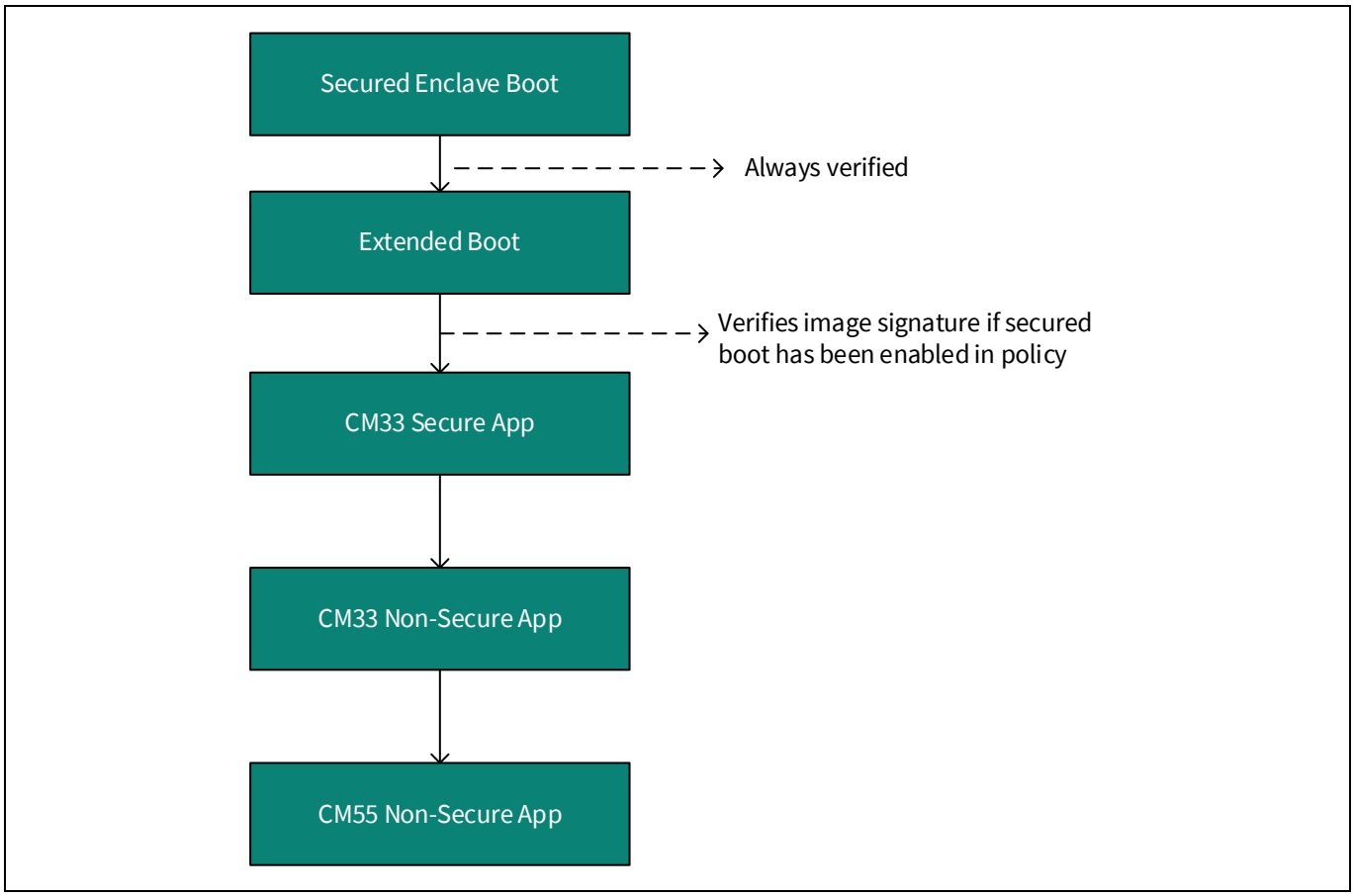

Figure 2 Chain of trust with Extended Boot

4.2.2 Chain of trust with extended boot and edge protect bootloader

The chain of trust with the Extended boot and Edge Protect Bootloader is as described in Figure 3.

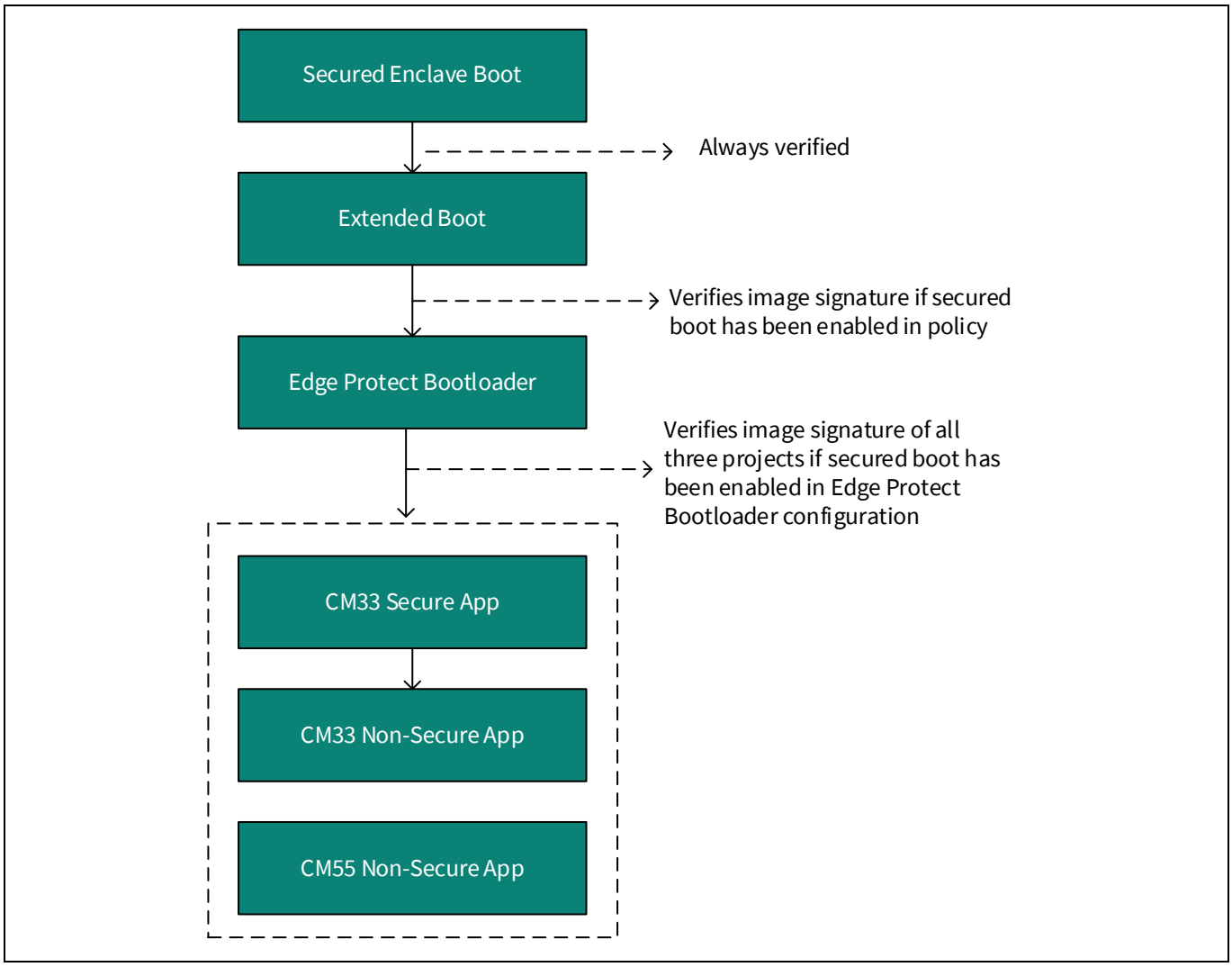

Figure 3 Chain of trust with edge protect bootloader

4.3 Hardware diagram

Figure 4 shows the hardware block diagram.

- **UART pins**
  - P6_7 (UART_TX)
  - P6_5 (UART_RX)
- GPIO LED pin
  - P16_7 (LED)

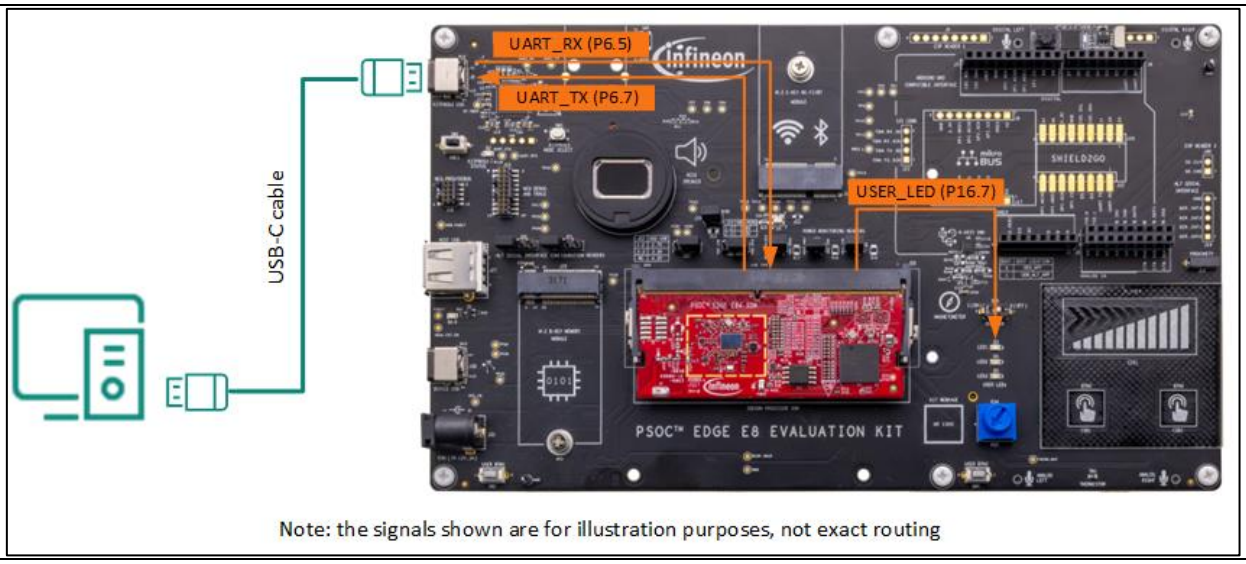

Figure 4 Hardware diagram

4.4 Project creation

1. Open the ModusToolbox™ Dashboard 3.6 and launch the Eclipse IDE for ModusToolbox™.

Note: If you have not installed ModusToolbox™, see the Required development tools section for more details.

Note:

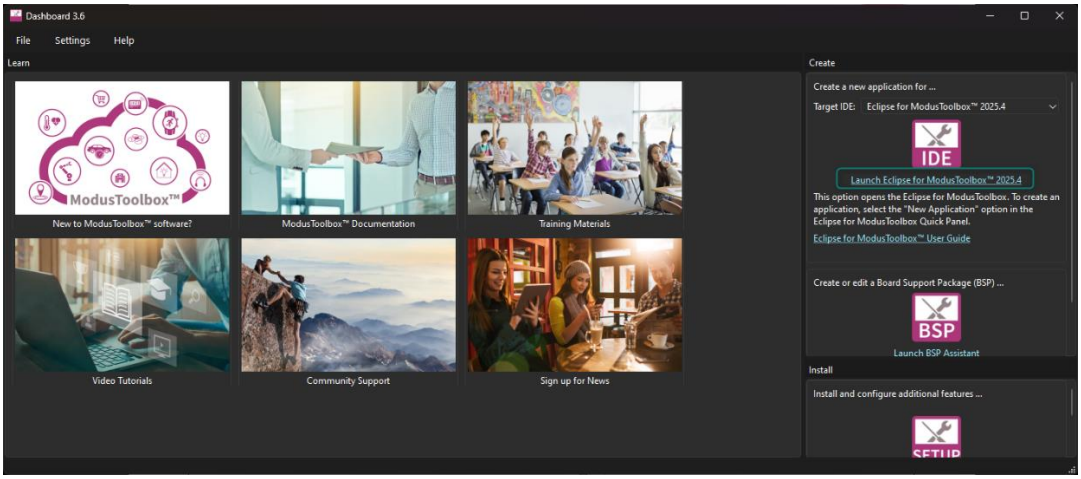

Figure 5 Open ModusToolbox™

2. Choose a **workspace directory** for your project. After selecting a suitable directory, click the **Launch** button.

Note: It is recommended to keep the workspace path short to avoid issues with access to files in Windows.

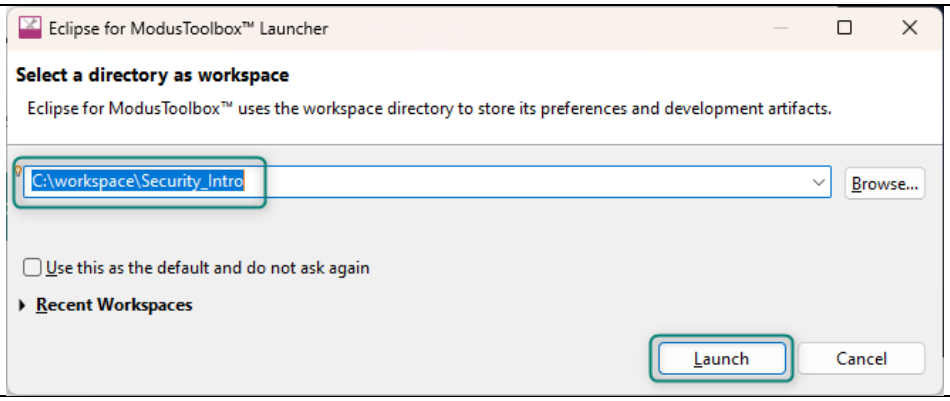

Figure 6 Choose workspace directory

3. Select **New Application** from the Quick Panel. Alternatively, go to **File> New> ModusToolbox™** application.

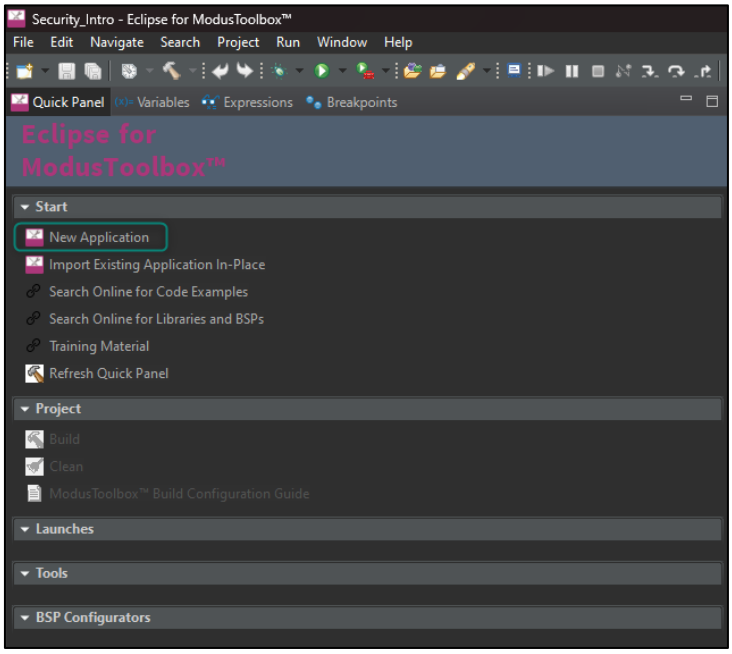

Figure 7 New application

4. Click New Application, which opens the **Project Creator Tool** and prompts you to choose a board support package (**BSP)**.

*Note: BSPs are aligned with our development/evaluation kits; they provide files for basic device functionality. A BSP typically has a **design. modus** file that configures clocks and other board-specific capabilities. That file is used by the ModusToolbox™ configurators. A BSP also includes the required device support code for the device on the board. You can modify the configuration to suit your application.*

5. Once **Projector Creator** launches, select the **KIT_PSE84_EVAL_EPC2 BSP** under the **PSOC™ Edge BSPs** and click **Next**.

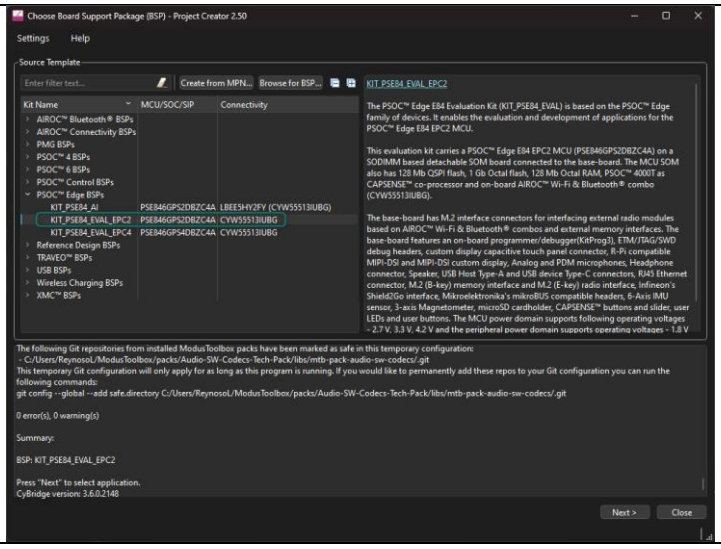

Figure 8 Project creator

Code example: Basic secure application

6. A new window opens to select an application. Select the **PSOC™ Edge Basic Secure App** under the Getting Started section. You can do this by selecting the **checkbox** near the application. Click **Create**.

Note: To prevent issues with Windows path length, it is recommended to rename the project if your worskpace path is long.

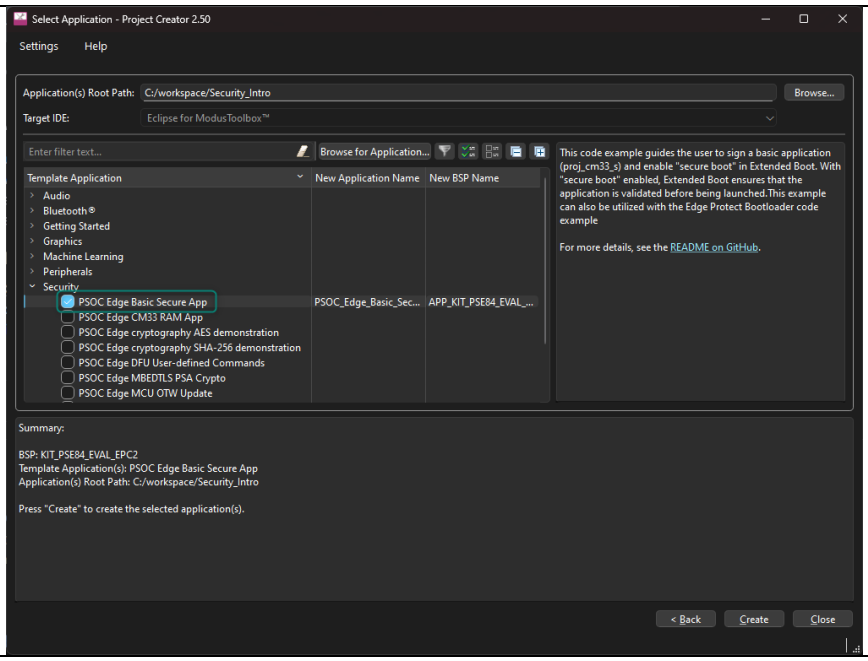

Figure 9 Select application

7. When you click **Create**, a new project is created, and you can see all the files in the Project Explorer window.

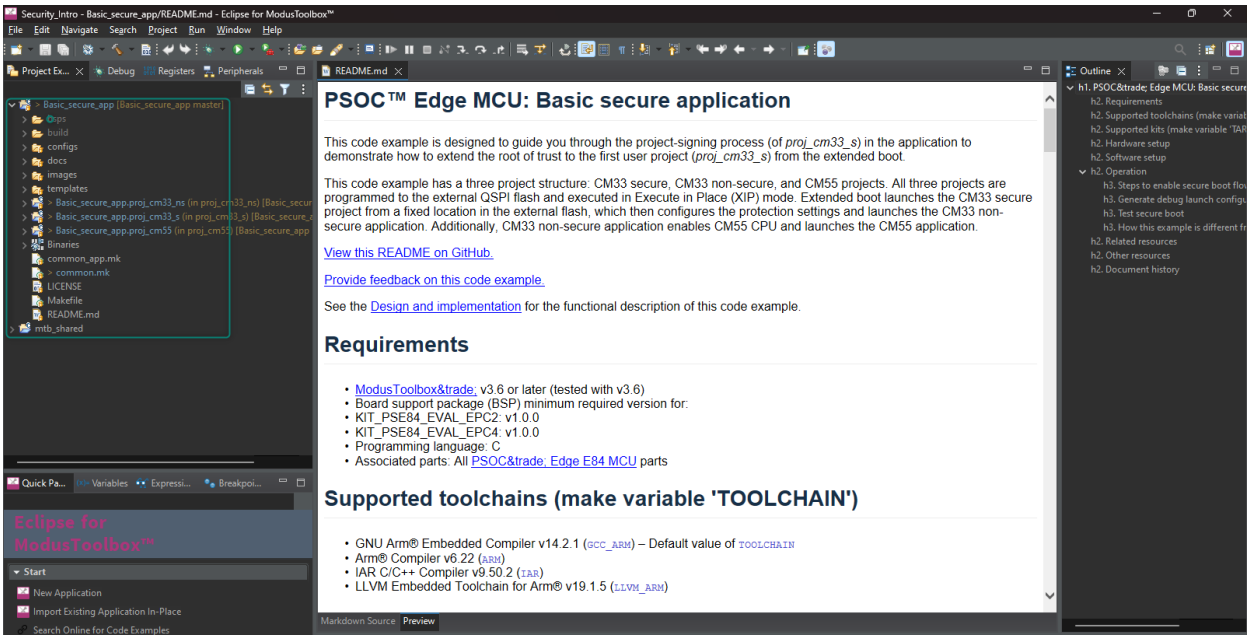

Figure 10 Creation of new application

8. Run the application by clicking on the **Run** option in the **Quick Panel**.

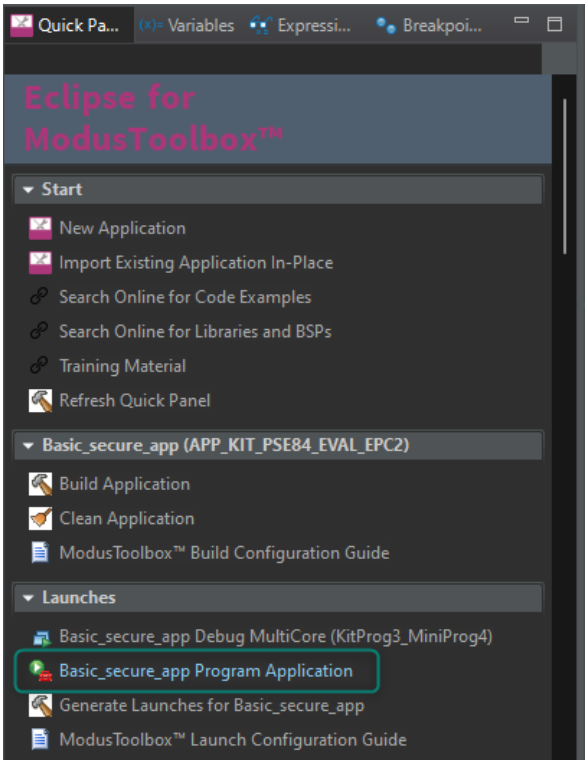

Figure 11 **Run the application**

4.5 Output

After programming is successful, open the serial terminal and configure at a baud rate of 115200-8-N-1. Then press the kit reset button. A message is printed as in the image below, and the kit LED blinks at approximately 1 Hz. Press **Enter** to pause and resume the LED blinking.

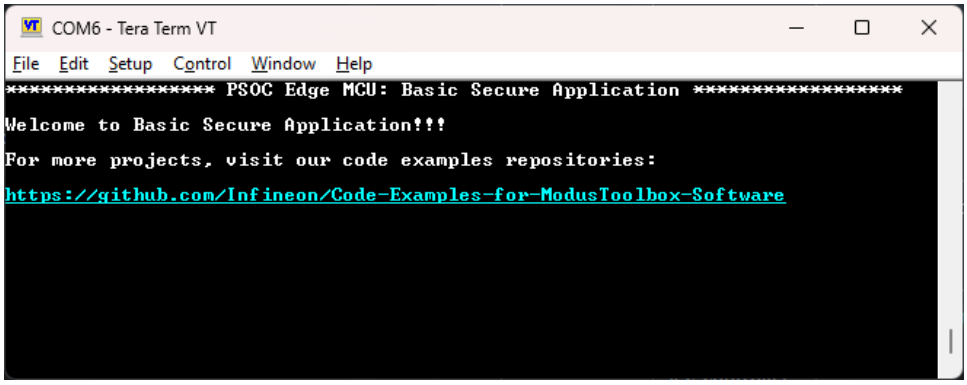

Figure 12 **Terminal output**

4.6 Ownership transfer

1. Edge Protect tools are packaged with **Edge Protect Security Suite v1.6**.
2. To use the executable, ensure that the executable path is added to your system environment "PATH".

   a) Open the Windows Control Panel and look for the option to **Edit the system environment variable.**

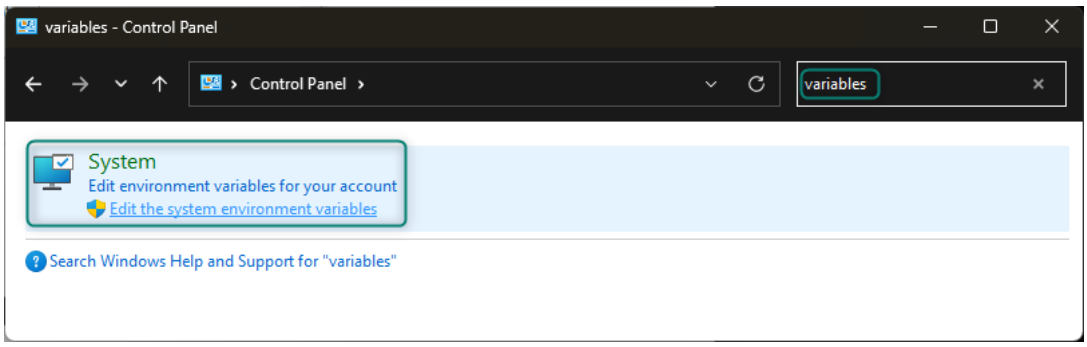

Figure 13 Open settings to change system environment variables

   b) Select the system variable **Path** and click **Edit**.

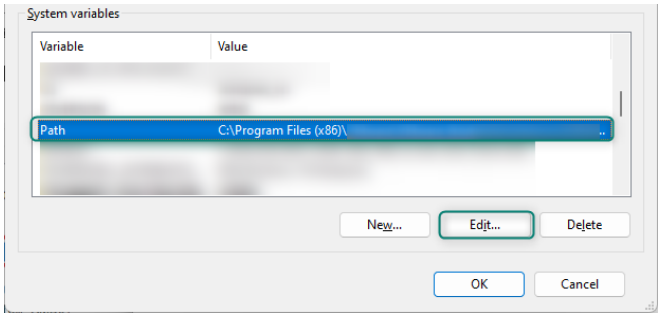

Figure 14 Modify the path variable

   c) Add a new path pointing to the Edge Protect Tools executable in your EAP. Click **OK**.

```
C:/Users/<username>/Infineon/Tools/ModusToolbox-Edge-Protect-Security-Suite-
1.6/tools/edgeprotecttools/bin/
```

3. Open modus-shell from the Windows Start Menu.
4. Check to ensure that you are using the **edgeprotecttools** executable from the EAP by executing the following:

```
$ whereis edgeprotecttools
```

*Note: Edgeprotecttools executable should be located at "C:/Users/<**username**>/Infineon/Tools/ModusToolbox-Edge-Protect-Security-Suite-1.6/tools/edgeprotecttools/bin/"*

5. Check the edgeprotecttools v1.6.0.XXXX

```
$ edgeprotecttools version
```

6. Navigate to your application folder

```
$ cd C:/workspace/Security_Intro/Basic_secure_app
```

7. Initialize Edgeprotecttools

```
$ edgeprotecttools -t pse8xs2 init
```

8. Configure the openocd path for edgeprotecttools

```
$ edgeprotecttools set-ocd --name openocd --path <openocd_path>
```

9. Replace <openocd_path> with the path to the openocd directory installed in ModusToolbox™ Programming tools v1.5.0 which is located at *C:\Infineon\Tools\ModusToolboxProgtools-1.5\openocd*

You need to create one key pair (private and public keys) using edgeprotecttools.

```
$ edgeprotecttools create-key --key-type ECDSA-P256 --output
keys/oem_private_key_0.pem keys/oem_public_key_0.pem
```

4.6.1 Create Certificate Signing Request (CSR)

The command below creates a CSR that can be sent to Infineon to create an Infineon-signed OEM certificate (OEM_CERT). For development, the Life Cycle Stage will be set to "**Development**".

```
$ edgeprotecttools -t pse8xs2 oem-csr --certificate-name "oem-cert" --oem
"Dummy OEM" --project "Dummy Project" --project-number "1234" --public-
key-0 keys/oem_public_key_0.pem --cert-type development --output
packets/apps/prov_oem/oem_csr.bin --sign-key-0 keys/oem_private_key_0.pem
```

4.6.2 Get the CSR signed from Infineon

10. Sign up for an Infineon Online Software Tools and Services (OSTS) account. Any developer may create an OSTS account by registering at osts.infineon.com.

11. Once you have registered, login to your OSTS account and click **Edge Protect Signing Service**.

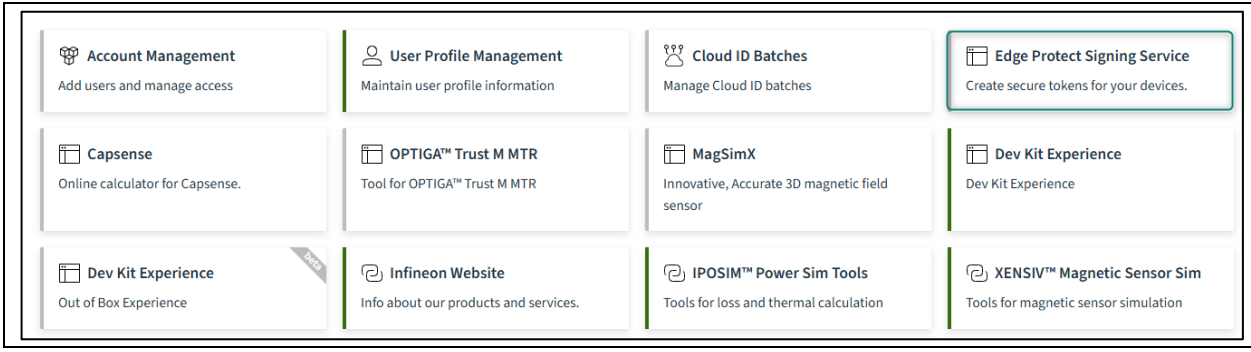

Figure 15 Open Edge Protect Signing Services

12. Click the Upload **New Certificate Request** button.

Figure 16 Create a new certificate request

13. Enter the certificate details:
    d) Enter the certificate name without any spaces or special characters in the **Name** field.
    e) Select "PSOC Edge E84 Rev.B0" as **MCU Revision**.
    f) Enter the description for this certificate in the **Description** field.
    g) Click the **Drop file here or click upload** button to upload the CSR that was generated in the previous steps and select the csr.bin file located at: *<application-directory>/packets/apps/prov_oem/oem_csr.bin*.
    h) Click **Submit**.

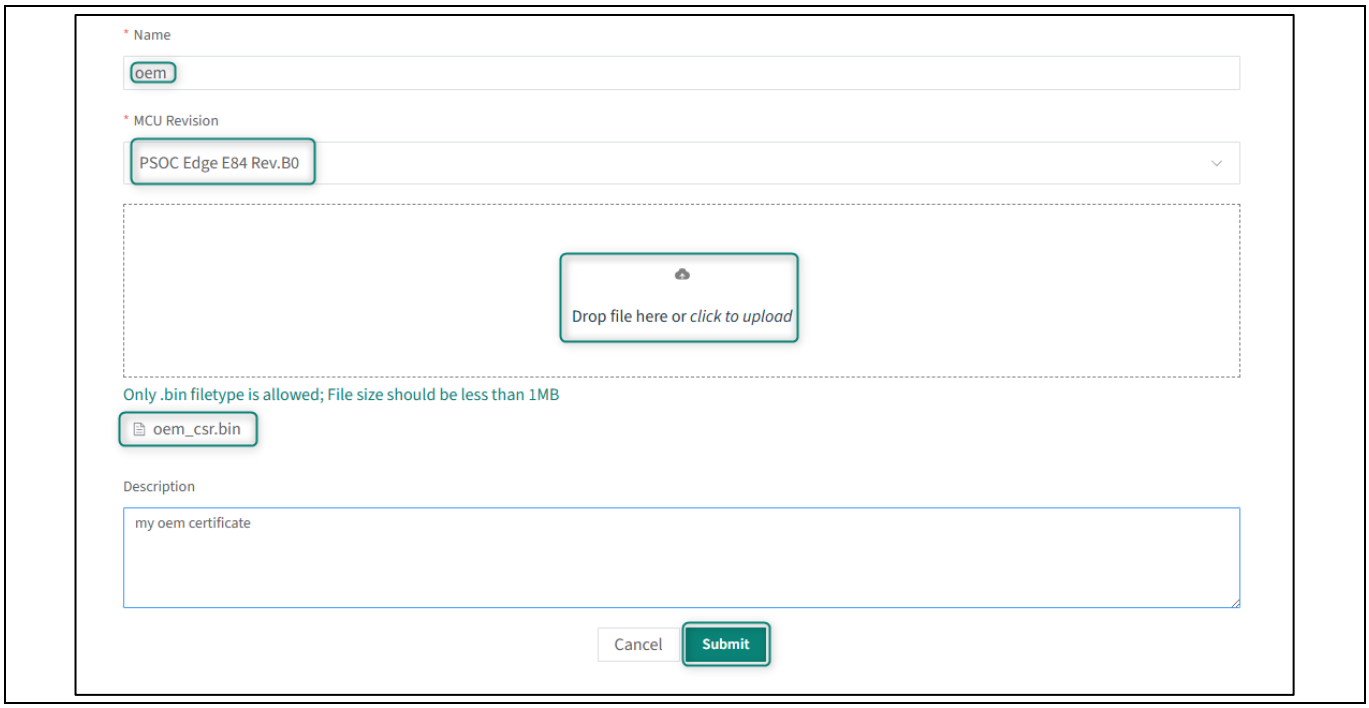

Figure 17 Enter certificate details

14. Download your signed certificate from the list of certificates under **Manage Certificate.** Creating the certificate might take a few minutes.

Name	Description	Created Date	MCU Revision	Status	Actions
oem	my oem certificate	2025-09-25	B0 Production	Created	Download Delete

Figure 18 Download certificate

15. The certificate will be downloaded as *_cert.bin format. For example, if the certificate name is given as "oem" the downloaded file will be "oem_cert.bin". Copy the downloaded certificate to the application inside the following directory *<application-directory>/packets/apps/prov_oem*.

4.6.3 Provision the device with your certificate and keys

16. In modus-shell, navigate to <application-directory> and execute the following command:

```
$ edgeprotecttools -t pse8xs2 provision-device -p
policy/policy_oem_provisioning.json --key keys/oem_private_key_0.pem --ifx-
oem-cert packets/apps/prov_oem/oem_cert.bin
```

Note: "—ifx-oem-cert" parameter is only required to change keys and certificate, and/or replace extended boot image. To only change the OEM policy configuration, this parameter can be skipped.

17. Ensure that provisioning is successful from the logs.

```
: 0 : INFO  : ** Detected Device: PSE846GPS2DBZC4A
: 0 : INFO  : ** SROM Boot version: 2.0.0.6022
: 0 : INFO  : ** RRAM Boot version: 2.0.0.7127
: 0 : INFO  : ** SE RT Services Base version: 1.0.0.2361
: 0 : INFO  : ** SE RT Services version: 0.0.0.0
: 0 : INFO  : ** Extended Boot version: 1.1.0.1700
: 0 : INFO  : ** Boot Status : CYBOOT_SUCCESS
: 0 : INFO  : ** Life Cycle  : DEVELOPMENT
: 0 : INFO  : Info : [cat1d.sys] Examination succeed
: 0 : INFO  : Info : gdb port disabled
: 0 : INFO  :     TargetName         Type       Endian TapName            State
: 0 : INFO  : --  ------------------ ---------- ------ ------------------ ------------
: 0 : INFO  :  0* cat1d.sys          mem_ap     little cat1d.cpu          halted
: 0 : INFO  : Info : Listening on port 6668 for tcl connections
: E : INFO  : Chip lifecycle stage: DEVELOPMENT
: E : INFO  : ROM Boot is ready for DLM package upload
: E : INFO  : Load application 'C:\workspace\Security_Intro\Basic_secure_app\packets\apps\prov_oem\cyapp_prov_oem_dlm.he
x'
: E : INFO  : Validation of the DLM package success
: E : INFO  : Start DLM package execution
: E : INFO  : Application execution completed with status code: (0xf2a00001) - CYAPP_SUCCESS: The provisioning applicati
on completed successfully
: E : INFO  : Chip lifecycle stage: DEVELOPMENT
: E : INFO  : ****************************************
: E : INFO  :              PROVISIONING PASSED
: E : INFO  : ****************************************
```

Figure 19 Successful provisioning

4.7 Enable Secure Boot in Extended Boot

1. Edit <application-directory>/policy/policy_oem_provisioning.json file using any text editor to set the value field to true for the “secure_boot” parameter. Save your changes after editing the file.

```
"extended_boot_policy": {
  "secure_boot": {
    "description": "Disable/Enable secure boot option by setting the value to false/true",
    "value": true
  },
```

Figure 20 Enabling secure boot in policy

2. In modus-shell, navigate to <application-directory> and execute the following command to provision the device with the updated policy.

```
$ edgeprotecttools -t pse8xs2 provision-device -p
policy/policy_oem_provisioning.json --key keys/oem_private_key_0.pem
```

3. Try resetting the EVK and note that the application won’t be executed.

Note: At this point the secure boot has been enabled in the extended boot, and hence the application will not boot after reset as it is not signed. Follow the steps below to enable post-build signing of the CM33 S image in the code example. Once the CM33 S signed image is programmed, the extended boot will be able to verify the signature and boot the application.

4.8 Enable signing of the first user application (cm33 s project)

1. In the MTB Basic Secure App code example, double-click on the common.mk under top top-level directory to open it for editing.

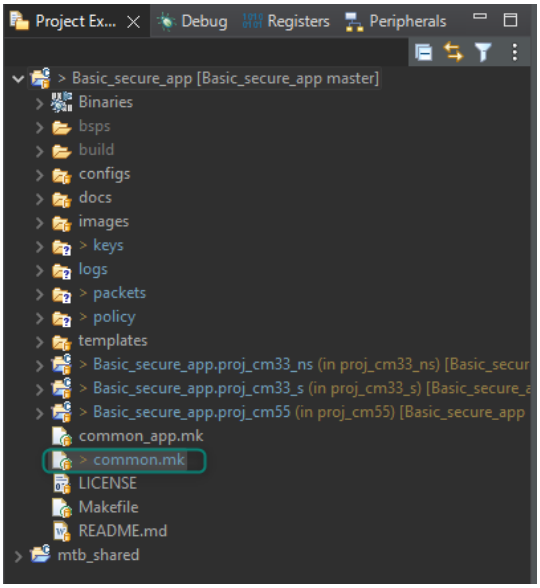

Figure 21 Open common.mk in the application

2. Change the JSON file input for the make variable “COMBINE_SIGN_JSON” to “secure_boot_with_extended_boot.json”. This JSON file contains the path to the key used for signing. Save the file after making changes

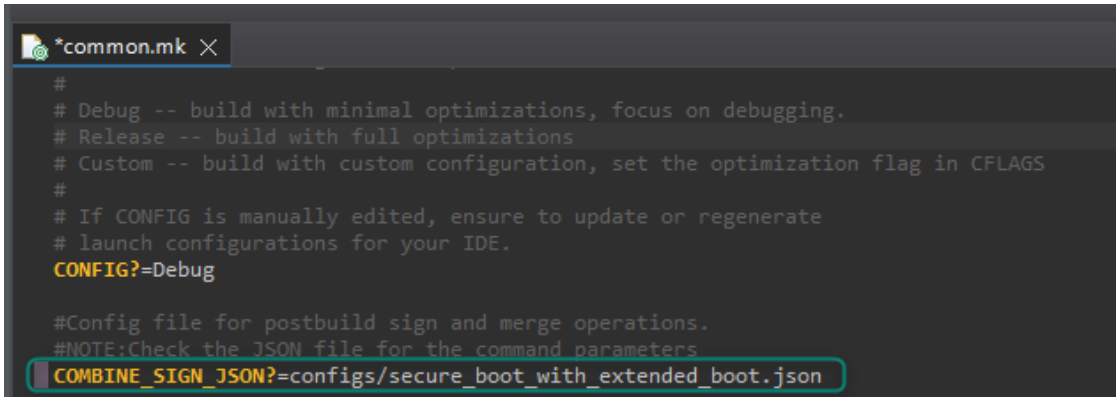

Figure 22 Edit the JSON file path in common.mk file

3. Build, program, and execute the application by clicking on the **Program Application** option in the Quick Panel.
4. Check the output on the terminal for the successful booting of verified application.

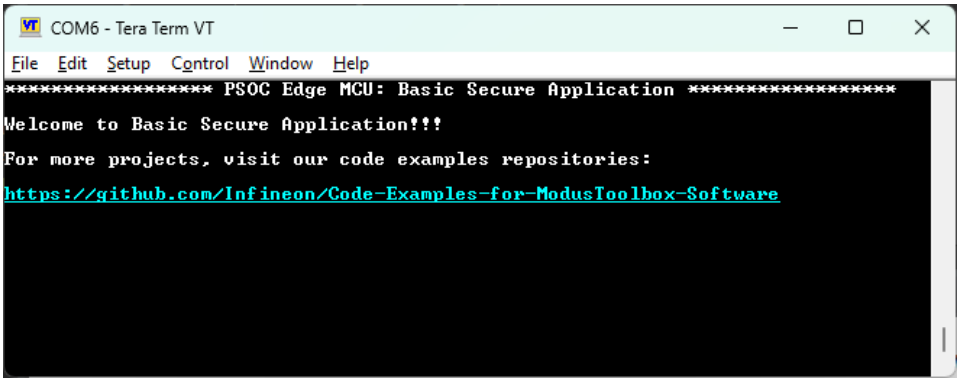

Figure 23 Terminal output after enabling Secure Boot

4.9 Conclusion

We have completed the following:

- Generated keys and certificate
- Transferred the ownership of the device
- Enabled ‘secure boot’ feature in the Extended Boot
- Launched a verified application

Note: (Try it yourself) Use wrong keys to sign the application and test whether the application will boot.

4.10 Challenge

Change the start address of the secure project (proj_cm33_s) to start from 0X70200000.

Suggestions:

- Modify start address using the Extended Boot policy file
- Use Memory Configurator to customise the memory map
- Update boot address in signer JSON file

4.11 Solution

1. Open the policy file and update the “oem_alt_app_address” in the Extended Boot policy to ‘0x70200000’.

```
"oem_alt_app_address": {
  "description": "Alternate starting location of OEM application",
  "value": "0x70200000"
},
```

Figure 24 Change oem_alt_app_address in policy file

2. Provision the device with the updated policy file.

```
$ edgeprotecttools -t pse8xs2 provision-device -p
policy/policy_oem_provisioning.json --key keys/oem_private_key_0.pem
```

3. Try resetting the EVK and note that the application won’t be executed. After provisioning, the extended boot will look for a valid user application at the address 0x702000000.
4. Open the device configurator tool by clicking “Device Configurator” in the Quick Panel.

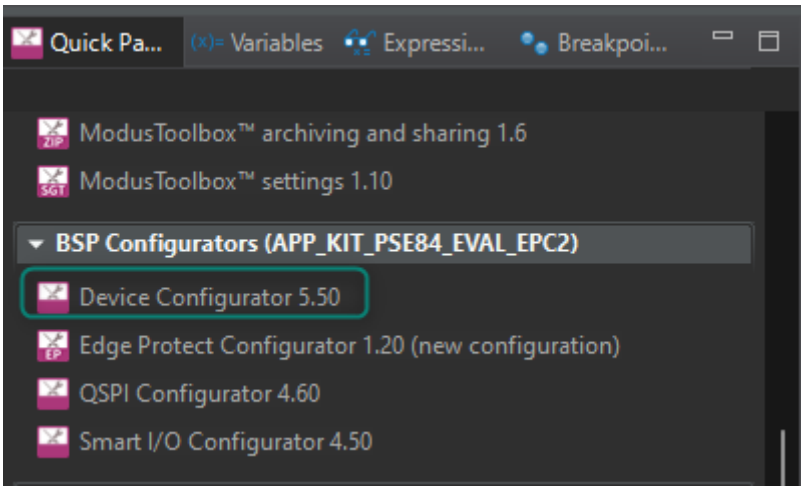

Figure 25 Open Device Configurator

5. In the device configurator window, open the memory tab and double-click click **m33s_nvm** region.

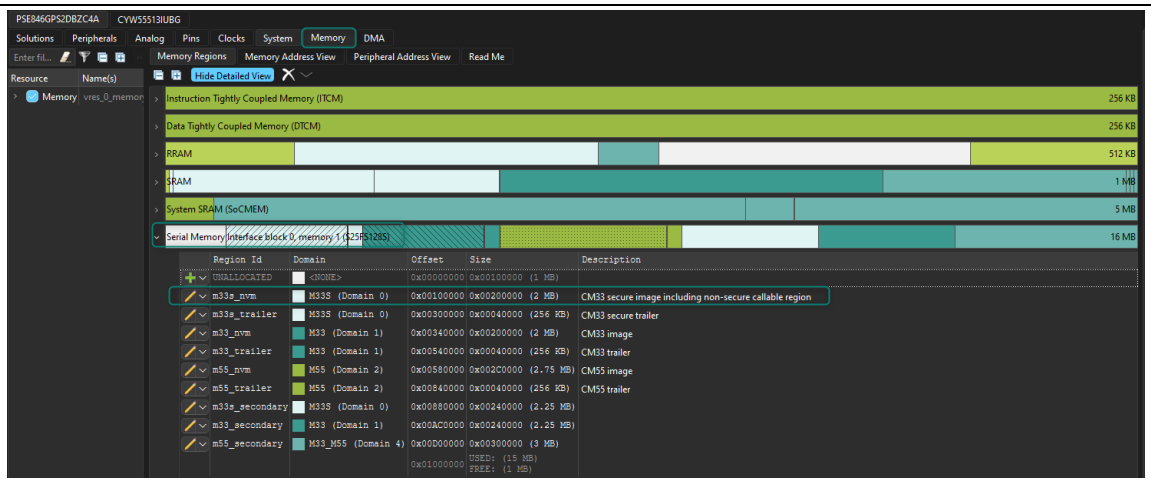

Figure 26 Select memory region in memory configurator

6. Edit the memory region start offset to start from “0x00200000” and adjust the image size so that it does not overlap with the next section. Click **OK**.

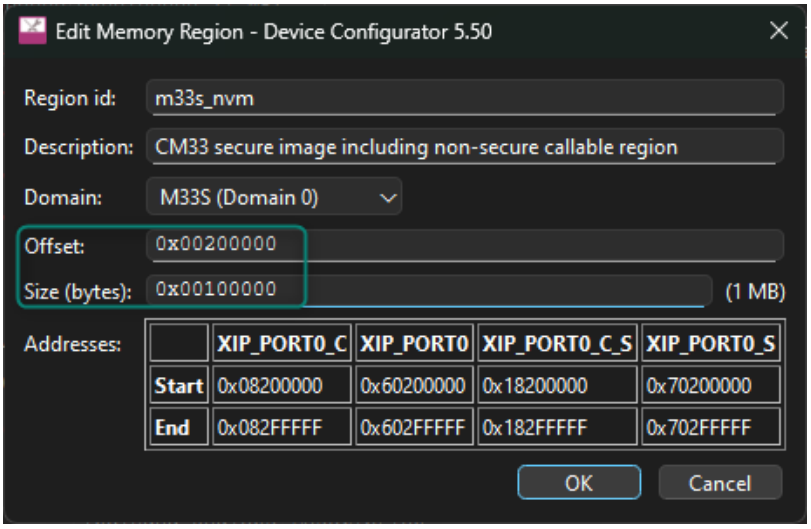

Figure 27 Adjust memory offset and size

7. Save the changes in the device configurator to regenerate files and close the tool.
8. If the cm33_s application contains veneer files, remove them. This is required because the address of the secure project was modified after the last build.

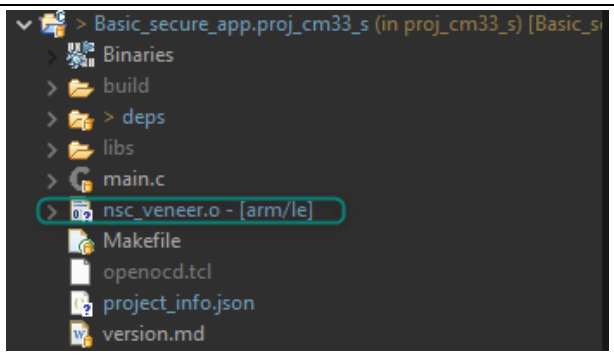

Figure 28 Remove veneer files from cm33_s

9. Build and run the application. You should observe that the application is executed again.

4.12 Revert to default configuration.

The changes made during this training may have an impact on the functionality of other code examples out the box. It is recommended to revert all changes, unless you are planning to continue working with a secure device and/or booting from a custom address.

1. Revert all changes made to the extended boot policy and provision the device again.
   i) Disable the Secure Boot.

```
"extended_boot_policy": {
  "secure_boot": {
    "description": "Disable/Enable secure boot option by setting the value to false/true",
    "value": false
  },
```

Figure 29 Disabling secure boot

   j) Revert back to the default value for oem_alt_app_address.

```
"oem_alt_app_address": {
  "description": "Alternate starting location of OEM application",
  "value": "0x70100000"
},
```

Figure 30 Revert oem_alt_app_address

   k) Save the policy file.
   l) Provision the device:

```
$ edgeprotecttools -t pse8xs2 provision-device -p
policy/policy_oem_provisioning.json --key
```

2. If the challenge was implemented, the Basic Secure App was modified to boot from a different location. If this modification is not required for further development, revert the changes in the memory configurator, or create a new copy of the Basic Secure App.

Revision history

Document revision	Date	Description of changes
**	2025-11-14	Initial release

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